

Air System Sizing Summary for Room AC Unit

(In Alternative: Default Alternative)

Project: HAP CEP

12/15/2025

Prepared by: RAC

9:31 PM

Air System Information

Air System Name **Room AC Unit**
Equipment Class **PKG ROOF**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **219.2** sqft
Location **Sialkot, Pakistan**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.5** Tons
Total coil load **18.4** MBH
Sensible coil load **18.0** MBH
Coil CFM at peak load **907** CFM
Sum of peak zone CFM **907** CFM
Sensible heat ratio **0.976**
CFM/Ton **591.6**
sqft/Ton **143.0**
BTU/(hr sqft) **83.9**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.5 / 73.0** F
Entering DB / WB **75.6 / 62.6** F
Leaving DB / WB **57.1 / 55.7** F
Resulting RH **50** %
Design supply temp. **58.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **13.4** MBH
Coil CFM at Design Heating **907** CFM
Max coil CFM **907** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **61.1**
Ent. DB / Lvg DB **69.4 / 83.3** F

Supply Fan Sizing Data

Design CFM **907** CFM
Design CFM/sqft **4.14** CFM/sqft

Fan motor BHP **0.50** BHP
Fan motor kW **0.39** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **19** CFM
CFM/sqft **0.09** CFM/sqft

CFM/person **17.00** CFM/person

Zone Sizing Summary for Room AC Unit

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Equipment Class PKG ROOF
Air System Type SZCAV

Number of zones 1
Floor Area 219.2 sqft
Location Sialkot, Pakistan

Sizing Calculation Information

Calculation Months Jan to Dec
Sizing Data Calculated

Zone CFM Sizing Sum of space airflow rates
Space CFM Sizing Individual peak space loads

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/sqft	Reheat Coil Capacity (MBH)	Reheat Coil Water gpm @ 20.0 F	Zone Htg Unit Coil Capacity (MBH)	Zone Htg Unit Water gpm @ 20.0 F	Mixing Box Fan Airflow (CFM)
Zone 1	907	907	4.14	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (sqft)
Zone 1	16.2	July 16:00	13.3	219.2

Space Loads and Airflows

Zone Name / Space Name	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (sqft)	Space CFM/sqft
Zone 1						
Room	16.2	July 16:00	907	13.3	219.2	4.14

Air System Heat Balance Summary for Room AC Unit

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Table 1. System Loads

COMPONENT LOADS	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.5 F / 73.0 F			OA DB / WB 39.4 F / 39.4 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Zone Conditioning	-	15981	283	-	14136	0
Plenum Load	-	0	0	-	0	0
Return Fan Load	907 CFM	0	-	907 CFM	0	-
Ventilation Load	19 CFM	611	73	19 CFM	597	0
Supply Fan Load	907 CFM	1344	-	907 CFM	-1344	-
Zone Fan Coil Fans Load	-	0	-	-	0	-
>> Total System Loads	-	17936	355	-	13389	0
Central Cooling Coil	-	17950	446	-	0	0
Central Heating Coil	-	0	-	-	13396	-
>> Total Conditioning	-	17950	446	-	13396	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Table 2. Zone Heat Balance Loads

Zone Heat Balance Component	DESIGN COOLING - JULY 16:00			DESIGN HEATING		
	OA DB / WB 106.5 F / 73.0 F			OA DB / WB 39.4 F / 39.4 F		
	Details	Sensible [BTU/hr]	Latent [BTU/hr]	Details	Sensible [BTU/hr]	Latent [BTU/hr]
Exterior Wall Convection	531 sqft	4407	-	531 sqft	4333	-
Roof Convection	219 sqft	7136	-	219 sqft	5219	-
Window Convection	17 sqft	495	-	17 sqft	402	-
Skylight Convection	0 sqft	0	-	0 sqft	0	-
Door Convection	52 sqft	691	-	52 sqft	458	-
Floor Convection	219 sqft	773	-	219 sqft	1290	-
Interior Wall Convection	0 sqft	0	-	0 sqft	0	-
Ceiling Convection	0 sqft	0	-	0 sqft	0	-
Overhead Lighting Convection	162 W	346	-	0 W	0	-
Task Lighting Convection	0 W	0	-	0 W	0	-
Electric Equipment Convection	164 W	421	-	0 W	0	-
People Convection	1	81	225	0	0	0
Infiltration	10 CFM	337	-14	11 CFM	373	0
Miscellaneous Equipment	-	0	0	-	0	0
Air Internal Energy Change	-	0	-	-	0	0
Safety Factor	10% / 5%	1468	11	10%	1208	0
>> Total Zone Loads	-	16153	221	-	13284	0
Key:	Positive values are cooling loads Negative values are heating loads			Positive values are heating loads Negative values are cooling loads		

Note 1: Surface convection line items show the combined effects of conductive heat gain to the surface and radiative heat gains absorbed at the surface which are then convected to room air.

Note 2: Lighting, equipment, and people line items include only the direct convective heat gain from the heat source to the room air. The radiative portion of the heat gain is first absorbed by surfaces in the room and then later convected from the surface to the air. Therefore the effect of the radiative portion of the heat gain is found in the surface convection line items.

Note 3: Solar heat gain is absorbed by surfaces in the room, re-radiated to other surfaces, and finally convected from the surfaces to room air. Therefore, the effect of solar heat gain is found in the surface convection line items.

System Psychrometrics for Room AC Unit

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DESIGN COOLING DAY AT JULY 16:00

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (F)	Specific Humidity (lb/lb)	Airflow (CFM)	CO2 Level (ppm)	Sensible Heat (BTU/hr)	Latent Heat (BTU/hr)
Ventilation Air	Inlet	106.3	0.01035	19	400	611	73
Vent - Return Mixing	Outlet	75.6	0.00952	907	772	-	-
Central Cooling Coil	Outlet	57.1	0.00941	907	772	17950	446
Central Heating Coil	Outlet	57.1	0.00941	907	772	0	-
Supply Fan	Outlet	58.5	0.00941	907	772	1344	-
Cold Supply Duct	Outlet	58.5	0.00941	907	772	-	-
Zone Air	-	75.0	0.00948	907	782	15981	283
Return Air	Outlet	75.0	0.00950	907	780	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1.080; At site altitude = 1.048 BTU/(hr-CFM-F)

Air Density x Heat of Vaporization x Conversion Factor: At sea level = 4746.6; At site altitude = 4604.8 BTU/(hr-CFM)

Site Altitude = 837.0 ft

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (BTU/hr)	T-stat Mode	Zone Cond (BTU/hr)	Zone Temp (F)	Zone Airflow (CFM)	CO2 Level (ppm)	Terminal Heating Coil (BTU/hr)	Zone Heating Unit (BTU/hr)
Zone 1	15978	Cooling	15981	75.0	907	782	0	0

System Psychrometrics for Room AC Unit

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WINTER DESIGN HEATING

TABLE 1: SYSTEM DATA

Component	Location	Dry-Bulb Temp (F)	Specific Humidity (lb/lb)	Airflow (CFM)	CO2 Level (ppm)	Sensible Heat (BTU/hr)	Latent Heat (BTU/hr)
Ventilation Air	Inlet	39.4	0.00523	19	400	-597	0
Vent - Return Mixing	Outlet	69.4	0.00523	907	400	-	-
Central Cooling Coil	Outlet	69.4	0.00523	907	400	0	0
Central Heating Coil	Outlet	83.3	0.00523	907	400	13396	-
Supply Fan	Outlet	84.7	0.00523	907	400	1344	-
Cold Supply Duct	Outlet	84.7	0.00523	907	400	-	-
Zone Air	-	70.0	0.00523	907	400	-14136	0
Return Air	Outlet	70.0	0.00523	907	400	-	-

Air Density x Heat Capacity x Conversion Factor: At sea level = 1.080; At site altitude = 1.048 BTU/(hr-CFM-F)

Air Density x Heat of Vaporization x Conversion Factor: At sea level = 4746.6; At site altitude = 4604.8 BTU/(hr-CFM)

Site Altitude = 837.0 ft

TABLE 2: ZONE DATA

Zone Name	Zone Sensible Load (BTU/hr)	T-stat Mode	Zone Cond (BTU/hr)	Zone Temp (F)	Zone Airflow (CFM)	CO2 Level (ppm)	Terminal Heating Coil (BTU/hr)	Zone Heating Unit (BTU/hr)
Zone 1	-14136	Heating	-14136	70.0	907	400	0	0